Churn Drivers and PCA Behavioural Axes — Group Report

Oct 27, 2026 · @Anish

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Answer in one line: yes to the first part — a regularised Logistic Regression matches a Random Forest (test ROC-AUC 0.842 vs 0.844) while exposing readable churn drivers — and partly to the second: PCA finds two stable, distinct axes, a *spend / service-breadth* axis (PC1) that is unrelated to churn (r = 0.01) and a *commitment / tenure* axis (PC2) that carries it (r = −0.51).

1. Research question and hypotheses

Research question. Can an interpretable supervised model identify churn drivers, and does PCA reveal independent behavioural axes? Data: IBM Telco Customer Churn, 7,043 customers, 19 business predictors (tenure, monthly and total charges, contract, billing, demographics, 9 service indicators), target `Churn` (26.5 % Yes).

Both hypotheses were written in the notebook before any model was trained, and each is checked by code rather than by eye.

- **H1 — interpretable is enough.** Logistic Regression (LR) reaches a test ROC-AUC within 0.02 of, or above, the Random Forest (RF), because the main drivers act roughly monotonically; both models have churn recall below 0.60 at threshold 0.5. **Verdict: supported.**
- **H2 — independent behavioural axes.** Fitted without labels, PC1 and PC2 load on different business variables (disjoint top-3 loadings); a 2-PC representation loses ⩾ 0.03 ROC-AUC against the original predictors, while the PCs reaching 90 % variance stay within 0.01. **Verdict: partly supported** — the axes are distinct, but 2 PCs lose only 0.015, less than predicted.

2. Data, leakage audit and protocol

The data are complete except 11 blank `TotalCharges`, all for customers with tenure 0 and none churners; we set them to 0 (not yet billed) by a row-wise rule that uses no dataset statistics. A median imputer would have charged these new customers about 1,400. A "never churns" rule already scores 73.5 % accuracy with zero churn recall, so accuracy is reported only as a reference.

Risk checked	Finding	Decision
Identifier	`customerID` unique per row	dropped
Target leakage	no column is a consequence of leaving (no cancellation date or retention offer)	19 predictors kept
Derived feature	`TotalCharges` ≈ tenure × `MonthlyCharges` (r = 1.00)	kept; the three coefficients are read jointly
Structural duplication	"No internet service" repeated in 6 add-on columns plus `InternetService = No`	tested in Experiment 2
Preprocessing leakage	imputer, encoder, scaler or PCA fitted on all rows would see the test set	all inside one `Pipeline`, refitted per fold
Temporal leakage	single snapshot: churners' tenure is their final tenure	cannot be fixed here; stated as a limitation

Fixed evaluation protocol. One stratified 80/20 split (seed 42; 5,634 train, 1,409 test, both 26.5 % churn). Every comparison uses the same frozen stratified 5-fold CV on the training set (mean ± SD). The test set is scored once per configuration, never used for a choice, and its hash is checked unchanged at three points in the notebook. Hyper-parameters are fixed a priori: LR with L2, C = 1; RF with 500 trees, `min_samples_leaf` = 10, `max_features` = sqrt.

Preprocessing pipeline. Encoding → median imputer (safety net) → `StandardScaler` on every column, so LR coefficients are per standard deviation and PCA sees a common scale. Two encodings, both label-free:

- **A — full one-hot:** each category of the 16 categorical variables gets its own 0/1 column (46 features).
- **B — compact business encoding:** Yes/No → 1/0; an add-on service → 1 only if subscribed; `Contract` → committed months (0, 12, 24); one-hot only for `InternetService` and `PaymentMethod` (22 features).

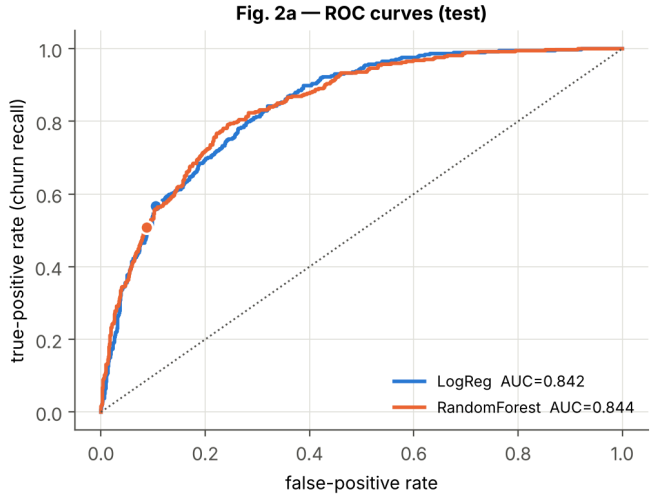
Metrics. ROC-AUC for overall ranking; PR-AUC (average precision, chance = 0.265) because it focuses on the minority churn class; churn recall and precision at threshold 0.5; MCC as a single-number summary of the confusion matrix. All results are appended to an experiment log (`results/experiment_log.csv`, 25 rows with configuration and seed).

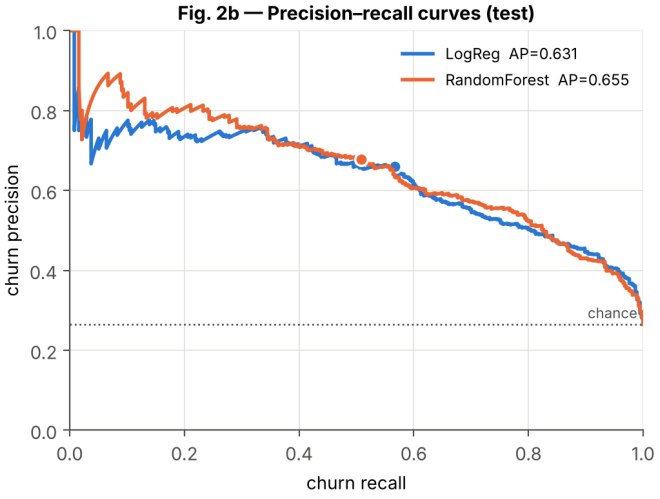
3. Results

Experiment 1 — Logistic Regression vs Random Forest

The two models are statistically indistinguishable on ranking: LR minus RF averages −0.001 ROC-AUC across the five paired folds, and LR wins 2 of 5. LR catches more churners at 0.5 (recall 0.567 vs 0.508) at slightly lower precision. H1 is supported.

Encoding A, threshold 0.5	CV ROC-AUC	CV PR-AUC	CV churn recall	Test ROC-AUC	Test PR-AUC	Test recall	Test precision	Test MCC
Logistic Regression	0.846 ± 0.014	0.661 ± 0.021	0.545 ± 0.047	0.842	0.631	0.567	0.660	0.486
Random Forest	0.848 ± 0.010	0.662 ± 0.022	0.491 ± 0.013	0.844	0.655	0.508	0.679	0.466
Never-churns baseline	—	—	—	0.500	0.265	0.000	0.000	0.000





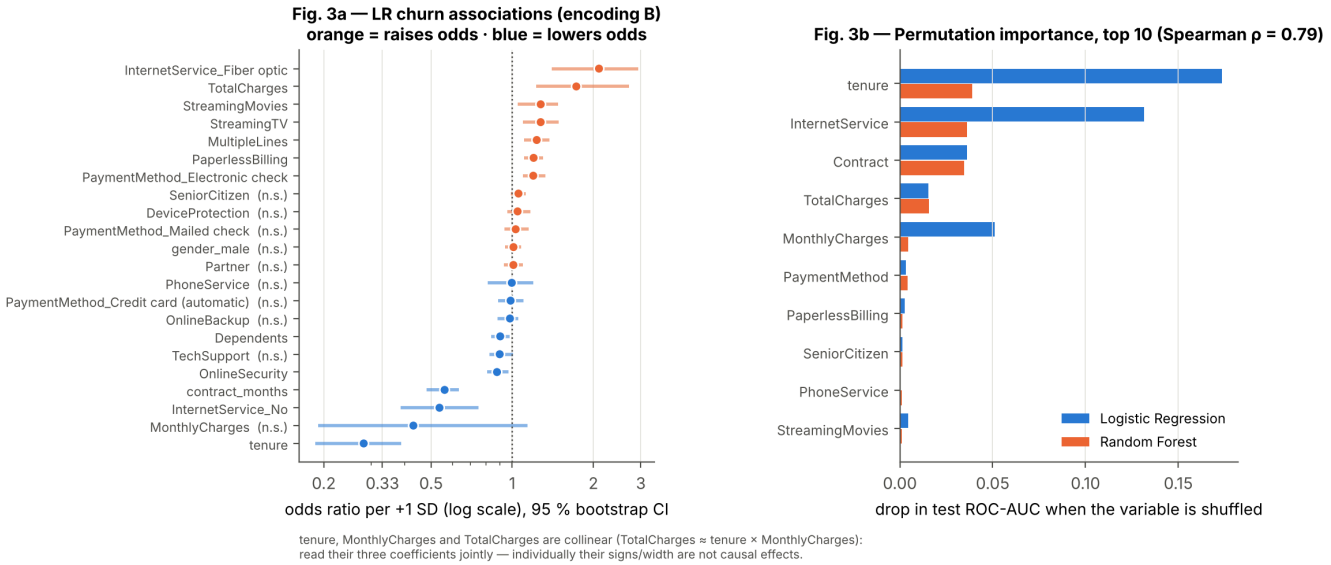
dots = operating point at threshold 0.5

Experiment 2 — two legitimate encodings, test set untouched

Encoding changes accuracy by nothing that matters (LR CV ROC-AUC 0.846 under both; RF 0.848 vs 0.847) but changes interpretability a lot. Under one-hot (A), the single fact "no internet" sits in 7 identical columns; L2 splits its weight evenly (−0.125 each), so no single coefficient means anything. Under the compact encoding (B), each business fact has one coefficient with a bootstrap 95 % CI (200 refits on the training set).

The clearest associations (odds ratio per +1 SD): tenure 0.28 [0.19–0.39], committed contract months 0.56 [0.48–0.63], no internet 0.54 [0.39–0.75], fibre optic 2.10 [1.40–2.93], electronic check 1.20 [1.09–1.33], paperless billing 1.20 [1.10–1.30]. `MonthlyCharges` (CI 0.19–1.14) and `TotalCharges` (1.73) must be read together with tenure because the three

are collinear. Permutation importance on the 19 original variables puts tenure, internet service and contract first for both models (Spearman ρ = 0.79 between LR and RF rankings), so the interpretable model and the black box agree on the drivers.



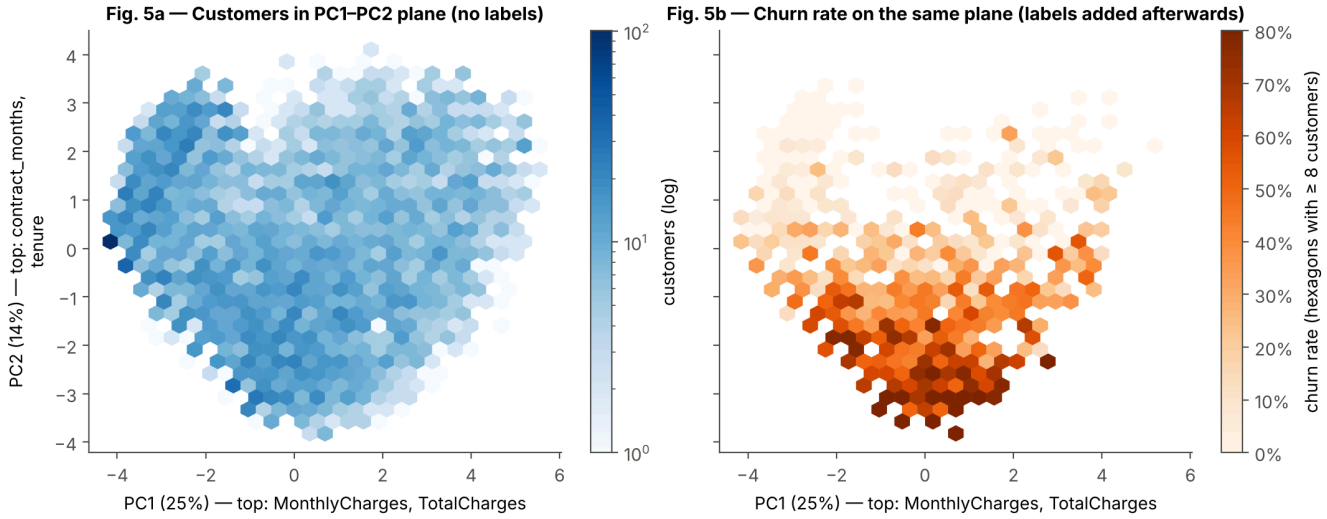
tenure, MonthlyCharges and TotalCharges are collinear (TotalCharges ≈ tenure × MonthlyCharges): read their three coefficients jointly — individually their signs/width are not causal effects.

Experiment 3 — PCA without labels

PCA on the 22 encoded, scaled training features gives PC1 = 25.1 % and PC2 = 14.1 % of variance; 15 components reach 92 %. PC1 is a **spend / service-breadth** axis (top loadings `MonthlyCharges`, `TotalCharges`, streaming), PC2 a **commitment** axis (contract months and tenure positive, electronic check and fibre negative). Their top-3 variables are disjoint.

Robustness: across 200 bootstrap resamples the loading vectors barely move (mean |cos| 1.00 for PC1, 0.999 for PC2). Across encodings they move more: the PC1 scores under A and B correlate at |r| = 0.73, PC2 at 0.68, so the axes depend on how categories are encoded.

Only after fitting did we add the labels: churn is unrelated to PC1 (r = 0.01) and strongly tied to PC2 (r = −0.51). Churn concentrates at low PC2 across the whole width of PC1 (Fig. 5).



Compression costs little for LR but more for RF. H2 is only partly supported: 2 PCs lose 0.015 ROC-AUC for LR, below the 0.03 we predicted.

CV, encoding B	LR ROC-AUC	LR PR-AUC	RF ROC-AUC	RF PR-AUC	LR test ROC-AUC
Original 22 features	0.846	0.662	0.847	0.664	0.842
PCA, 2 components	0.831	0.628	0.817	0.602	0.822
PCA, 15 components (92 %)	0.844	0.658	0.833	0.623	0.837

4. Interpretation and limitations

Why is churn recall not sufficient by itself? Recall rises just by flagging more people. The same LR reaches recall 0.95 at threshold 0.10, but it then flags 62 % of all test customers and only 41 % of flags are real churners (517 false alarms out of 1,409). Recall ignores the cost of every wrongly targeted customer, so we report it with precision, PR-AUC and the share flagged; the threshold itself is a business choice that weighs the cost of an offer against the value of a retained customer.

LR threshold (test)	Churn recall	Precision	Customers flagged
0.10	0.95	0.41	62 %
0.35	0.71	0.54	35 %
0.50	0.57	0.66	23 %
0.65	0.31	0.75	11 %

What is lost when using PCA rather than business variables? Actionability and units: a coefficient on "PC2" cannot be turned into an action such as "offer a one-year contract", because each component mixes all 22 standardised inputs. PCA maximises variance in the predictors, not churn signal — PC1 explains the most variance yet is unrelated to churn. The axes also depend on encoding ($|r| \approx 0.7$ between encodings A and B), and RF loses 0.030 ROC-AUC with 2 PCs because the rotation breaks the clean splits it uses on single variables.

How do we avoid treating correlation as causation? Odds ratios and importances describe associations in observational data. Month-to-month contracts, fibre and electronic check co-occur with short tenure, so they confound one another, and customers choose their contract themselves. We phrase every finding as "associated with". A causal claim needs an intervention, such as a randomised retention offer or contract incentive, or at least an explicit causal model (a DAG with adjustment or matching) validated on later data.

Limitations.

- One cross-sectional, synthetic IBM sample from a single (fictional) operator: results may not transfer to real customers or markets.
- Snapshot data: tenure and charges are measured when churn is observed, so a deployed model scoring active customers would see different distributions (temporal leakage risk).
- No time-ordered validation, no customer-interaction data (complaints, support calls, price changes), which are often stronger signals.
- Fixed, untuned hyper-parameters and one split seed; tuning or other seeds could move the small differences reported.

5. Individual contributions and oral defense

Each member owns one result and reproduces one parameter change from notebook §8; seeds stay fixed, so any difference comes from that change alone.

Member	Contribution	Result explained at the defense	Live change and expected effect
Pinninti Sai Anish	Data inspection, leakage audit, evaluation protocol, preprocessing pipeline, report assembly	Why accuracy (73.5 % for "never churns") and recall alone mislead	`THRESHOLD = 0.35` → recall 0.57 → 0.71, precision 0.66 → 0.54, AUCs unchanged

Member	Contribution	Result explained at the defense	Live change and expected effect
Paravasthu Vamsi Krishna Suri	Experiment 1 (LR vs RF, metric choice) and Experiment 2 (encodings, odds ratios, permutation importance)	LR ≈ RF on ROC-AUC (0.842 vs 0.844), and why one-hot splits the "no internet" coefficient 7 ways	`LR_C = 0.05` → coefficients shrink, ranking of drivers stays
Kola Karan	Experiment 3 (PCA, robustness checks, original vs reduced predictors)	PC1 = spend (r = 0.01 with churn), PC2 = commitment (r = −0.51)	`PCA_VAR_TARGET = 0.80` → fewer PCs, larger AUC loss

Reproducibility. `telco_churn_pca.ipynb` runs top to bottom in about 2 minutes with `SEED = 42`; `pytest tests/` executes it twice and confirms the experiment log is identical, all six figures exist and every metric beats the baseline.